# Tran The Duong

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M.S. student in Electrical Engineering at KAIST, focusing on multi-object tracking and robotic control. Builds perception models, control algorithms, and simulation software in Python, C++, and MATLAB.

## Education

**KAIST**
*M.S. in Electrical Engineering* | *Daejeon, South Korea* Sep. 2024 – Present
Expected graduation: Feb. 2027. Advisor: Prof. Dong Eui Chang.

**Hanoi University of Science and Technology**
*B.S. in Automatic Control* | *Hanoi, Vietnam* Sep. 2019 – May 2024
GPA: **3.89/4.00**. Thesis advisor: Assoc. Prof. Dao Phuong Nam.

## Research Experience

**Graduate Researcher, CtrlLab, KAIST**
*Computer vision, robotics, and control* Sep. 2024 – Present
*Current focus: Multi-object tracking (Oct. 2025 – Present).*

**BeliefCO** *Master's thesis, ongoing*
*End-to-End Transformer-Based Multi-Object Tracking for Drone and Bird Surveillance*

- Developed a Transformer-based tracker that combines motion uncertainty, object-existence estimates, and probabilistic association to maintain object identities.
- Created WOSDETC tracking annotations and recording-level data splits; evaluated the tracker on four aerial-tracking benchmarks.
- Achieved test HOTA scores of **72.32 on BFT**, **62.91 on WOSDETC**, and **65.77 on BSB**.

**Bayesian Motion Prompting** *Python, PyTorch*

- Extended MOTIP with uncertainty-aware motion predictions for identity matching; built training and evaluation workflows for DanceTrack, SportsMOT, and bird flock tracking.

## Robotics and Control Projects

**Wheeled-Bipedal Robot Modeling and Control**
*KAIST* | *MATLAB/Simulink, CasADi, C++, Gazebo* Sep. 2024 – Oct. 2025

- Modeled the robot as a wheeled inverted pendulum and formulated velocity-tracking optimization with soft Lyapunov constraints and state and motor-torque limits.
- Built a C++ interface between Gazebo and Simulink to exchange joint states, IMU measurements, and torque commands for controller development in simulation.

**Risk-Sensitive Stochastic Control (RS-JiLQG)**
*KAIST graduate course project* | *Python, NumPy* Sep. 2024 – Oct. 2025

- Co-developed a risk-sensitive iterative LQG formulation to control nonlinear systems under Gaussian noise and sudden wheel-slip disturbances.
- Implemented the solver and evaluated risk settings through Monte Carlo simulations of a wheeled inverted pendulum, comparing mean and entropic costs.

**PX4 Offboard Position Control with ROS 2**
*Python, ROS 2, PX4* Nov. 2024

- Implemented a ROS 2 node that sends position commands to PX4 and monitors vehicle state and command acknowledgments.

## Additional Experience

**Engineering Intern, Viettel Aerospace Institute**
*Embedded systems software* Sep. 2023 – Feb. 2024

- Designed user interfaces for embedded systems; worked with C++, MATLAB, and Simulink.

**Undergraduate Researcher, HUST**
*Reinforcement learning and control* 2022 – 2024

- Investigated output-feedback adaptive dynamic programming (ADP) and reinforcement learning control for a two-wheeled inverted pendulum in the undergraduate thesis.
- Studied ADP-based control for mobile robots with wheel slip and wind turbines with disturbance observers; designed a PID controller for a resistance furnace.

## Technical Skills

**Programming:** C, C++, Python | **Deep learning:** PyTorch.
**Modeling and simulation:** MATLAB, Simulink.
**Robotics and control:** Robot dynamics, optimal control, and state estimation.
**Reinforcement learning:** Strong foundational knowledge.
**Languages:** Vietnamese (native), English.

## Scholarships and Awards

- **KAIST Scholarship**, M.S. program.
- **Best Graduation Thesis**, HUST thesis defense council, Feb. 2024.
- **Academic excellence scholarships**, HUST, six semesters (2020–2023).
- **Corporate scholarship for outstanding students**, 2023.

## Additional Programming Projects

**C++ Programming Exercises**
*C++, Git* Apr. 2025
Maintained introductory C++ examples for language practice and version control.

**Pattern Recognition with RBM Features and Neural Classification**
*Python, NumPy, TensorFlow | KAIST coursework* Dec. 2024
Implemented a restricted Boltzmann machine feature extractor and neural classifier for MNIST, with sampling, normalization, and separate training, validation, and test sets.